

Cloning a die using a weak triple-minimum family

Research proof and exact nine-die certificate

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Theorem 1. *Let W be a three-wise fair word on n letters, each occurring M times, and distinguish a letter X . For a face of any other letter D , let h count occurrences of X before that face. Suppose*

$$\sum_{\text{faces of } D} h^2 = M^3/3 \quad (D \neq X). \quad (1)$$

Let $\rho_1, \dots, \rho_M$ be permutations of k new labels such that in every triple of labels, each label is earliest in exactly $M/3$ of the permutations. Replace occurrence r of X by the palindrome $\rho_r \rho_r^{\text{rev}}$, and replace every occurrence of every other letter by two consecutive copies. The resulting word is three-wise fair on $n - 1 + k$ letters, each occurring $2M$ times.

Proof. Call the k new labels clones. Each replacement palindrome contains two occurrences of every clone and is pairwise fair: both orders of any two clones have two scattered occurrences. A concatenation of h such blocks is again pairwise fair, and each order of two clones occurs $2h^2$ times.

A triple containing at most one clone inherits its order counts from the corresponding triple of W , multiplied by 2^3 . Now consider two clones A, B and one old letter D . Pairwise fairness of W gives $\sum h = M^2/2$, while (1) gives $\sum h^2 = M^3/3$. For each original face of D there are two copies in the new word. The number of occurrences of the order ABD is therefore

$$4 \sum h^2 = 4M^3/3 = (2M)^3/6.$$

The same holds for BAD . The two orders with D first follow from $\sum (M - h)^2 = M^3/3$. Each order with D in the middle has count

$$8 \sum h(M - h) = 8(M^3/2 - M^3/3) = 4M^3/3.$$

It remains to check three clones. In a single block $\rho \rho^{\text{rev}}$, a prescribed order ABC occurs zero times if B is earliest among its three labels in ρ , and twice otherwise. The within-block contribution over all M blocks is therefore $4M/3$. Occurrences using three distinct blocks contribute $8 \binom{M}{3}$ to each prescribed order. Occurrences using exactly two blocks contribute $8 \binom{M}{2}$: the doubled block may contain the first two or last two letters, each contributing $2 \cdot 2 = 4$. Thus every order has total count

$$8 \binom{M}{3} + 8 \binom{M}{2} + 4M/3 = 4M^3/3,$$

as required. □

Apply this with $M = 6$, distinguished label $X = 1$, and the four-letter seed

$$W = 012332103102201321033012.$$

The seed is three-wise fair. Its prefix-1 counts at the faces of the other letters are

D	$(h_1, \dots, h_6)$	$\sum h_i$	$\sum h_i^2$
0	(0, 2, 3, 3, 5, 5)	18	72
2	(1, 1, 3, 3, 4, 6)	18	72
3	(1, 1, 2, 4, 5, 5)	18	72.

Use the following six permutations on six new labels:

$$012345, \quad 521340, \quad 531204, \quad 413250, \quad 423105, \quad 032145.$$

For each of the twenty triples, each member is earliest in exactly two permutations. After cloning, relabel the three surviving old labels 0, 2, 3 as 6, 7, 8. The resulting word is exactly the certificate `../partial_fairness/nine_dice_twelve_three_wise.json`.

Corollary 1. *The minimum common face count for three-wise fair dice is $M_3(n) = 12$ for every $5 \leq n \leq 9$.*

Proof. The construction gives nine dice with twelve faces; delete labels for the smaller values of n . The independent six-face enumeration and mod-four extension obstruction in `six_face_extension_congruence.tex` rule out five six-face dice, and hence rule them out for all $n \geq 5$. Three-wise fairness requires $6 \mid M^3$, equivalently $6 \mid M$. Therefore no smaller positive common face count is possible. $\square$

The verifier `verify_nine_dice_cloning.py` uses only the Python standard library. It reconstructs the word, checks the seed and weak-minimum family, and directly enumerates all $\binom{108}{3}$ index triples of the final word. Every one of the $9 \cdot 8 \cdot 7 = 504$ ordered triples of distinct labels occurs exactly $12^3/6 = 288$ times. This verification uses no floating-point arithmetic or solver output.